

CORESTAR ANALYSIS

SETUP GUIDE

Six steps to build a setup you can trust before you look at a single tube.

1 LOAD THE DATA

Open the exam data and the calibration file. Confirm the probe type, frequencies, and tube range match the inspection plan.

CHECK Cal standard ID matches the data sheet



2 SET UP CHANNELS

Assign differential and absolute channels for each frequency. Build any required mixes (for example, support suppression) per your procedure.

CHECK Every channel is labeled and in the right order



CAL STD

3 SET THE PHASE

On the cal standard, rotate so probe motion is horizontal. Set the through-wall hole signal to the angle your procedure calls for.

CHECK Signal travels in the expected direction

4 SET THE VOLTAGE

Normalize the reference signal (for example, the 4 × 20% flat-bottom holes) to the voltage in your procedure. Apply it to all channels.

CHECK Reference reads the target voltage

5 BUILD THE CURVE

Mark each cal flaw (100%, 60%, 40%, 20%) and save the phase-to-depth curve. Points should form a smooth line.

CHECK No point sits far off the curve



6 SAVE & VERIFY

Save the setup with a clear name. Open a known tube and confirm supports and landmarks look normal before you start.

CHECK Setup name includes unit, probe & date



QUICK TIPS

Change one setting at a time and re-check the cal standard after each change.

Use the same setup for every tube in the same exam group.

If a step won't line up, stop and ask. Don't force it.

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HOW TO ANALYZE

A repeatable routine for every tube: scan, spot, measure, size, code, and report.

1 SCAN THE TUBE

Scroll the strip chart along the full length of the tube. Find the landmarks first: tube sheet, supports, and U-bend.

CHECK All expected landmarks are present



2 SPOT SIGNALS

Look for anything that breaks the normal background or looks different from a typical support signal.

CHECK Review the primary channel first, then the others



3 READ THE LISSAJOUS

Box the signal and look at its shape in the Lissajous window. Measure the peak-to-peak voltage and note the phase angle.

CHECK Signal is clean, not noise or probe wobble

4 SIZE IT

Read the depth from your calibration curve using the phase angle. Confirm the call on a second frequency or mix.

CHECK Both frequencies tell the same story

5 CODE & LOCATE

Give it the right report code (for example, % depth, DNT, or NDD). Record the location in inches from the nearest landmark.

CHECK Code comes from the approved code list



6 REPORT & REVIEW

Enter the result, save, and move to the next tube. Flag anything unusual for a second analyst or resolution review.

CHECK Every tube has a result, even if it's NDD



QUICK TIPS

When unsure, compare the signal to the same spot on a clean tube.

Always measure location from the nearest landmark, not the tube end.

If you can't call it with confidence, flag it. Don't guess.

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INDICATION FIELD GUIDE

The flaws we see most, which channel to call them on, and what to watch for.

IDI**ID pitting**

CH1 • M1 ON SUPPORTS

**LOOKS LIKE**

Small, sharp, clean figure-8. Angle sits on the ID side of the through-wall line.

TELL-TALE

Strongest on the highest frequency, since CH1 is most sensitive at the inside surface.

WATCH OUT

Probe noise, scratches, and ID deposits. Confirm it repeats on another channel.

ODI**OD pitting**

CH1 • M1 ON SUPPORTS

**LOOKS LIKE**

Figure-8 rotated past the through-wall line toward OD. Read depth from the cal curve.

TELL-TALE

Also shows on lower frequencies. Shallow OD pits can be weak on CH1 but clear on CH3/CH5.

WATCH OUT

Support edges and deposits pull the angle. Very low frequency drift alone is not an ODI.

WAR**Wear**

M2

**LOOKS LIKE**

A broad, slow signal right at a support or AVB once the mix removes the support.

TELL-TALE

Always at a support edge. Often shows on neighboring tubes in high-flow areas.

WATCH OUT

Normal mix residual can look like wear. Compare with clean supports in the same tube.

ERO**Erosion**

CH6 (ABSOLUTE)

**LOOKS LIKE**

A gradual drift on the absolute strip chart over a length of tube, not a sharp figure-8.

TELL-TALE

Common at the inlet end, just past the tube sheet. Differentials may barely see it.

WATCH OUT

Probe drift and the tube sheet transition. Compare with a clean stretch of the same tube.

DNT**Dent**

CH1 • M1 ON SUPPORTS

**LOOKS LIKE**

Large signal, lying close to the probe-motion direction. A geometry change, not wall loss.

TELL-TALE

High voltage with no flaw-like angle. Often at supports where corrosion squeezes the tube.

WATCH OUT

A dent can hide a flaw. Check inside it on the higher frequencies and the mix.

**Other codes to know**

USE YOUR APPROVED CODE LIST

NDD

No detectable degradation. Every clean tube still gets a result.

RST

Restriction. The probe won't pass, so record where it stopped.

PLG

Plugged tube. Not tested, but still recorded.

DSI

Distorted support signal. Flag it for a closer look.

Codes vary by site. Swap these for the ones in your procedure.

QUICK TIPS

At a support, switch to the mix before you call anything.

Call on the listed channel, then confirm on one other.

Not sure what it is? Flag it. Don't force a code.